CHEMICAL ENGINEERING ANALYSIS

Assignment 1

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CHG 433: Chemical Engineering Analysis

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ATTESTATION

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By the signatures above, each author attests to the fact that the contents of this paper are, to the best of our knowledge, neither plagiarised nor false.

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EXECUTIVE SUMMARY

Mathematical modelling and analysis is very important in Chemical Engineering. Engineers must design specialised reactors and processes to create new substances or materials while adhering to strict safety regulations and working to maximise efficiency from both an economic and ecological standpoint. The aim of this paper was to establish a firm knowledge in analysing, modelling and refining new and existing systems, establishing the balance between complexity and accuracy and foster team-work between members towards creating or advancing industrial processes. Four questions solved in four sections were evaluated using various numerical methods and the computer application software SCILAB and Microsoft Excel. The solutions were first computed manually, by making assumptions and applying theories. Then the required formulas and lines of code were typed into the software. Problem 1 depicted a chemical engineering process application involving 3 chemical reactors operating at steady state. By making material balance around each reactor, a mass continuity system of three simultaneous equations was generated. The equations were used to evaluate the concentrations of the required specie in each reactor: x_A, x_B, and x_C by solving the using Crammer's Rule, Gaussian Elimination, and Matrix Inverse. The Matrix inverse method was used in solving the system of equations on SCILAB and Microsoft Excel. The values of x_A, x_B, and x_C were gotten as 14.67mg/m^3, 7.33mg/m^3 and 10.10mg/m^3 respectively. Question 2 was in two parts. Part A dealt with the determination of the temperature in a Resistance Temperature Detector (RTD) operation. Using the Newton-Raphson and Interval Bisection methods, the temperature of the detector when its resistance was 300Ω was calculated as 260.22 ºC. The Excel and SCILAB solutions also employed the Newton-Raphson equation to calculate the required temperature. In part B, the volume of a real gas was determined at different pressures by using the Van der Waals equation. The problem was solved in SCILAB with the "fsolve" function, and in Excel using "Goal Seek". Pressure was then plotted against volume and plotted in the Excel sheet.

The plot showed an inverse relationship between pressure and volume (Boyle's Law). A model was derived for a liquid tank system with inlet and outlet volumetric flow rates in question 3. The model related the volumetric inlet flow to the level of liquid in the tank. The model derived was a lumped parameter model and the initial steady-state liquid level in the tank was determined to be 25cm after making a key assumption. An expression for the dynamic response of the liquid level to change in operation commands was also determined by linearising the non-linear model equation. This was done using Taylor Series Expansion, and the resulting ODE was evaluated with Laplace Transforms. The solution of the resulting equation was taken to SCILAB to show the response of the system to different step lengths 0.01, 0.05, 0.1, 0.5, 1, 10 by using the Euler Explicit function and Runge-Kutta 4th order methods. In question 4, the method of least-squares was used to best fit a function of the form $In(k) = C + bIn(T) - \frac{D}{T}$ to the data given. This data showed the measurements of the rate coefficient, k for a reaction at different temperatures, T. By manually developing a system of three simultaneous equations using least squares and Crammer's rule, the values C, b, and D were obtained as -30.365, -1.408x10^{-8}, and -1584.2 respectively, resulting in the equation: $In(k) = -30.365 - 1.408 * 10^{-}8In(T) - \frac{1584}{T}$. The values computed manually varied from the ones gotten in the software due to approximation and possible random errors. From comparing the function and constants gotten to the log of the Arrhenius Equation, $k = AT^{b}.e^{-\frac{E_a}{RT}}$, the values of A and E_a were deduced as 13171.04 J/mol and 6.496x10^{-8} using Excel and SCILAB. The work done is significant because it establishes the chemical engineering undergraduate in fundamental principles of solving process problems.

TABLE OF CONTENT

Attestation ..... **Error! Bookmark not defined.**

Executive Summary ..... **Error! Bookmark not defined.**

Table of Contents 6

Introduction ..... **Error! Bookmark not defined.**

SECTION 1 9

1.1 Manual Calculation 10

1.1.1 Cramer's rule 12

1.1.2 Matrix Inverse Method 13

1.1.3 Gaussian Elimination Method 14

1.2 Software Calculation 15

Analysis Description 15

SECTION 2 17

2.1 Manual Calculation 18

Part A 18

2.2 Software Calculation 20

Part A 20

Part B 21

SECTION 3 24

3.1 Manual Calculation 25

3.2 Software Calculation 29

SECTION 4 32

4.1 Manual Calculation 33

4.2 Software Calculation 37

Discussion ..... **Error! Bookmark not defined.**

Section 1 39

Section 2 40

Section 3 41

Section 4 42

Things We Learnt 43

Conclusion ..... **Error! Bookmark not defined.**

INTRODUCTION

Mathematical techniques and models are used in almost every element of a chemical engineer's professional life. Modern industrial production is based on solid scientific methods, many of which are part of chemical engineering. Engineers must design specialised reactors and processes to create new substances or materials while adhering to strict safety regulations and working to maximise efficiency from both an economic and ecological standpoint. Many innovations in material design and process development are thought to be fuelled by mathematical methods in chemical engineering.

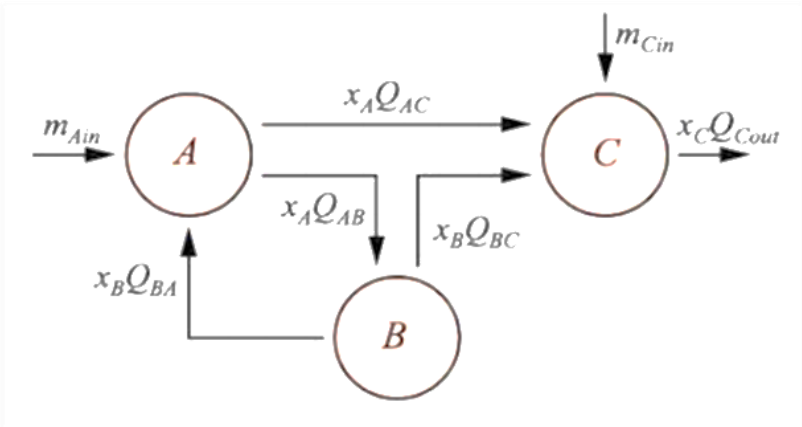
According to the American Institute of Chemical Engineers AIChE, Mathematical modelling is a powerful tool for understanding, designing, and predicting processes and process equipment in the chemical industry, including the conservation of momentum, energy, and material. Models are designed to predict and optimise the behaviour of engineering operations. These techniques and models provide the mathematical foundations of different areas of chemical engineering and describes typical applications. Modern chemical engineers are not only expected to be adept at using these techniques; they are also expected to be able to apply computer-based solutions when analytical ones are impractical.

In this paper, four potential situations are presented and analysed both manually and with the help of software. This paper aims to

- Establish a firm knowledge in analysing, modelling and refining new and existing systems.
- Give insight into predicting mathematical representations of real-life operations and establishing the balance between complexity and accuracy.
- Foster team-work between members towards creating or advancing industrial processes.

SECTION 1

The first consideration for this paper is in the setup below:



The chemical engineering process application given by the figure above involves three chemical reactors A, B, and C. At steady state, the concentrations of a particular specie in each reactor has the values x_A, x_B, and $x_C$in units of mg/m3. If the flow rates from reactor i (A, B, or C) to reactor j (A, B, or C) is denoted as Q_{ij} (units of m3/s), then the mass flow rate of species n from reactor i to reactor j is x_iQ_{ij} (units of mg/s). Since this chemical species is conserved (i.e., neither produced nor destroyed) conservation of mass (of the species) for each reactor must hold. For the process shown in the figure, Q_{AB} = 40 m^3/s, Q_{AC} = 80 m^3/s, Q_{BA} = 60 m^3/s, Q_{BC}= 20 m^3/s, Q_{Cout} = 150 m^3/s, mC_{in} = 195 mg/s, and mA_{in} = 1320mg/s. Write down the mass continuity equations for each reactor and solve them to find the concentrations x_A, x_B, and $x_C$in each reactor.

(a) Using only your calculator, compute the solutions manually using

  (i) Cramer's rule

  (ii) Matrix Inverse

  (iii) Gaussian elimination

(b) Solve the same problem using the (i) Excel Spreadsheet, (ii) SCILAB

1.1 Manual Calculation

The mass continuity equations are gotten from the material balances around each reactor.

The general material balance equation in a system is given by:

$\text{Input}_{\text{of material}}$ - $\text{Output}_{\text{of material}}$ + $\text{Generation}_{\text{of material}}$ - $\text{Consumption}_{\text{of material}}$ = $\text{Accumulation}_{\text{of material}}$

Assumptions

- Since the operation is at Steady State, Accumulation = 0
- No Generation and Consumption

Given: $Q_{\text{AB}} = 40\, m^3/s,\ Q_{\text{AC}} = 80\, m^3/s,\ Q_{\text{BA}} = 60\, m^3/s,\ m_{\text{Ain}} = 1320 mg/s.$

$$Q_{BC} = 20\, m^3/s, \qquad Q_{cout} = 150\, m^3/s, \qquad m_{Cin} = 195\, mg/s$$

Reactor A material balance

$$m_{\text{Ain}} + x_B\, Q_{\text{BA}} = x_{\text{A}}\, Q_{\text{AC}} + x_{\text{A}}\, Q_{\text{AB}}$$

$$m_{\text{Ain}} + x_{\text{B}}\, Q_{\text{BA}} = x_{\text{A}}\ (Q_{\text{AC}} + Q_{\text{AB}})$$

$$1320 + 60\, x_{\text{B}} = (80 + 40)\, x_{\text{A}}$$

$$\mathbf{120}\boldsymbol{x}_{\mathbf{A}} - \mathbf{60}\boldsymbol{x}_{\mathbf{B}} = \mathbf{1320} \quad \text{---1.1}$$

Reactor B material balance

$$x_{\text{A}} Q_{\text{AB}} = x_{\text{B}}\, Q_{\text{BA}} + x_{\text{B}}\, Q_{\text{BC}}$$

$$x_A(40) = x_B(60) + x_C(20)$$

$$\mathbf{40}\boldsymbol{x_A} - \mathbf{80}\ \boldsymbol{x_B} = \mathbf{0} \quad \text{---1.2}$$

Reactor C material balance

$$m_{\text{Cin}} + x_A Q_{\text{AC}} + x_B Q_{\text{BC}} = x_C Q_{\text{cout}}$$

$$195 + x_{\text{A}}\,(80) + x_{\text{B}}\,(20) = x_{\text{C}}\,(150)$$

$$195 + 80\,x_{\text{A}} + 20\,x_{\text{B}} = 150\,x_{\text{c}}$$

$$\mathbf{80\,x_A + 20\,x_B - 150\,x_C = 195} \quad \text{---1.3}$$

Equations 1-3 are presented below:

$$\mathbf{120x_{\text{A}} - 60x_{\text{B}} = 1320} \quad \text{----------1.1}$$

$$\mathbf{40x_A - 80\,x_B = 0} \quad \text{----------1.2}$$

$$\mathbf{80\,x_A + 20\,x_B - 150\,x_C = 195} \quad \text{----------1.3}$$

The mass continuity expressions in equations 1.1-1.3 can be expressed in Matrices of the form $Ax = B$ to solve for the matrix x.

The coefficient matrix is given by:

$$\text{A} = \begin{bmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{bmatrix}$$

The variable matrix is given by:

$$\text{x} = \begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix}$$

The constant matrix is given by:

$$\text{B} = \begin{bmatrix} 1320 \\ 0 \\ 195 \end{bmatrix}$$

1.1.1 Cramer's rule

$x_A = \frac{Dx_A}{D}$ ----------1.4a

$x_B = \frac{Dx_B}{D}$ ----------1.4b

$x_C = \frac{Dx_C}{D}$ ----------1.4c

Where D is the determinant of the coefficient matrix

$Dx_{A,B,C}$ are matrices of x_A, x_B, and x_C as defined below.

$$x_A = \begin{bmatrix} 1320 & -60 & 0 \\ 0 & -80 & 0 \\ 195 & 20 & -150 \end{bmatrix} \quad x_B = \begin{bmatrix} 120 & 1320 & 0 \\ 40 & 0 & 0 \\ 80 & 195 & -150 \end{bmatrix} \quad x_C = \begin{bmatrix} 120 & -60 & 1350 \\ 40 & -80 & 0 \\ 80 & 20 & 195 \end{bmatrix}$$

$$Dx_A = |Ax_A| = \begin{vmatrix} 1320 & -60 & 0 \\ 0 & -80 & 0 \\ 195 & 20 & -150 \end{vmatrix}$$

= 1320 [(-80 x -150) - (20 x 0)] – (-60) [(0 x -150) – (0 x 195)] + 0 [(0 x 20) – (-80 x 195)]

= 15840000

$$Dx_B = |Ax_B| = \begin{vmatrix} 120 & 1320 & 0 \\ 40 & 0 & 0 \\ 80 & 195 & -150 \end{vmatrix} = 7920000$$

$$Dx_C = |Ax_C| = \begin{vmatrix} 120 & -60 & 1350 \\ 40 & -80 & 0 \\ 80 & 20 & 195 \end{vmatrix} = 10908000$$

$$D = |A| = \begin{vmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{vmatrix} = 1080000$$

∴ $\boldsymbol{x_A}$= 15840000 / 1080000 = **14.67 mg/m^3**

$\boldsymbol{x_B}$ = 7920000 / 1080000 = **7.33 mg/m^3**

$\boldsymbol{x_C}$ = 1080000 / 1080000 = **10.10 mg/m^3**

1.1.2 Matrix Inverse Method

The mass continuity expressions expressed in Matrices of the form Ax + b = C to solve for the matrix x.

$$Ax = B$$

$$\begin{bmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{bmatrix} \begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix} = \begin{bmatrix} 1320 \\ 0 \\ 195 \end{bmatrix} \text{-------1.5}$$

$$x = A^{-1} B$$

$$\begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix} = \begin{bmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{bmatrix}^{-1} \begin{bmatrix} 1320 \\ 0 \\ 195 \end{bmatrix}$$

$The\ inverse\ of\ a\ function\ \ A^{-1} = \frac{1}{Determinant}$ x Adjunct A

Adjunct A = Transpose of the Cofactor A

$$\text{Cofactor A} = \begin{bmatrix} +\begin{vmatrix} -80 & 0 \\ 20 & -150 \end{vmatrix} & -\begin{vmatrix} 40 & 0 \\ 80 & -150 \end{vmatrix} & +\begin{vmatrix} 40 & -80 \\ 80 & 20 \end{vmatrix} \\ -\begin{vmatrix} -60 & 0 \\ 20 & -150 \end{vmatrix} & +\begin{vmatrix} 120 & 0 \\ 80 & -150 \end{vmatrix} & -\begin{vmatrix} 120 & -60 \\ 80 & 20 \end{vmatrix} \\ +\begin{vmatrix} -60 & 0 \\ -80 & 0 \end{vmatrix} & -\begin{vmatrix} 120 & 0 \\ 40 & 0 \end{vmatrix} & +\begin{vmatrix} 120 & -60 \\ 40 & -80 \end{vmatrix} \end{bmatrix}$$

$$\text{Cofactor A} = \begin{bmatrix} 12000 & 6000 & 7200 \\ -9000 & 18000 & -7200 \\ 0 & 0 & -7200 \end{bmatrix}$$

$$\text{Adjunct A} = \begin{bmatrix} 12000 & -9000 & 0 \\ 6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}$$

$$\text{Inverse} = \frac{1}{1080000} \begin{bmatrix} 12000 & 9000 & 0 \\ -6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{90} & -\frac{1}{120} & 0 \\ -\frac{1}{180} & \frac{1}{60} & 0 \\ \frac{1}{150} & -\frac{1}{150} & -\frac{1}{150} \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix} = \begin{bmatrix} \frac{1}{90} & -\frac{1}{120} & 0 \\ \frac{1}{180} & \frac{1}{60} & 0 \\ \frac{1}{150} & -\frac{1}{150} & -\frac{1}{150} \end{bmatrix} \begin{bmatrix} 1320 \\ 0 \\ 195 \end{bmatrix} = \begin{bmatrix} \mathbf{14.67} \\ \mathbf{7.33} \\ \mathbf{10.10} \end{bmatrix}$$

$\boldsymbol{x_A}$**= 14.67 mg/m³,** $\boldsymbol{x_B}$ **= 7.33 mg/m³,** $\boldsymbol{x_C}$ **= 10.10 mg/m³**

1.1.3 Gaussian Elimination Method

The mass continuity equations can be presented in row echelon form below.

$$\left[\begin{array}{ccc|c} 120 & -60 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & 195 \end{array}\right]$$

Then the new row $R_1 \rightarrow \frac{80}{60} R_1$

$$\left[\begin{array}{ccc|c} 160 & -80 & 0 & 1760 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & 195 \end{array}\right]$$

Then the new row $R_1 \rightarrow R_1 - R_2$

$$\left[\begin{array}{ccc|c} 120 & 0 & 0 & 1760 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & 195 \end{array}\right]$$

From this, the values of the concentrations can be evaluated by substitution from the equations below:

$$120\, \boldsymbol{x_A} = 1760 \qquad \text{----------1.6a}$$

$$40\, \boldsymbol{x_A} - 80\, \boldsymbol{x_B} = 0 \qquad \text{----------1.6b}$$

$$80\, \boldsymbol{x_A} + 20\, \boldsymbol{x_B} - 150\, \boldsymbol{x_C} = 195 \qquad \text{----------1.6c}$$

$$\boldsymbol{x_A} = \frac{1760}{120} = \mathbf{14.67\ mg/m^3},$$

$$\boldsymbol{x_B} = \frac{\mathbf{40}(\boldsymbol{x_A})}{\mathbf{80}} = \mathbf{7.33\ mg/m^3}$$

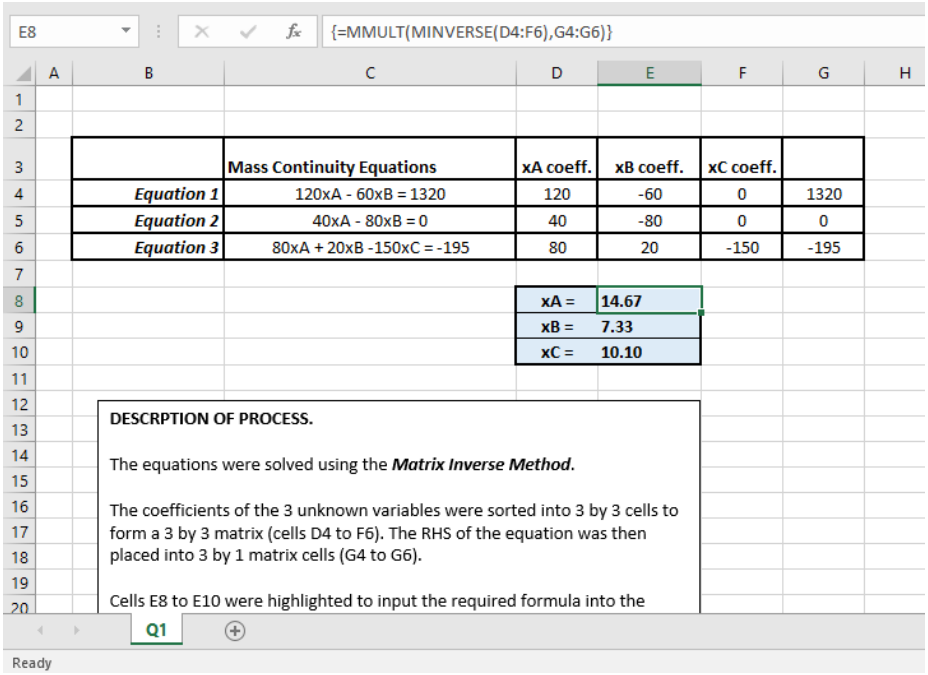
$$\boldsymbol{x_C} = \mathbf{10.10\ mg/m^3}$$

1.2 Software Calculation

Analysis Description

On Excel, the equations were solved using the ***Matrix Inverse Method***.

- The coefficients of the 3 unknown variables were sorted into 3 by 3 cells to form a 3 by 3 matrix (cells D4 to F6).
- The RHS of the equation was then placed into 3 by 1 matrix cells (G4 to G6).
- Cells E8 to E10 were highlighted to input the required formula into the formula bar.
- The "**MMULT**" function was then used to multiply the inverse of the 3x3 matrix, *D4:F6* (using the "**MINVERSE**" function) by the 3x1 matrix, *G4:G6* to get the values of x_A, x_B, and x_C.
- The formula used was: **=MMULT(MINVERSE(D4:F6),G4:G6)**



E8 {=MMULT(MINVERSE(D4:F6),G4:G6)}

	Mass Continuity Equations	xA coeff.	xB coeff.	xC coeff.	
Equation 1	120xA - 60xB = 1320	120	-60	0	1320
Equation 2	40xA - 80xB = 0	40	-80	0	0
Equation 3	80xA + 20xB -150xC = -195	80	20	-150	-195

xA =	14.67
xB =	7.33
xC =	10.10

DESCRPTION OF PROCESS.

The equations were solved using the ***Matrix Inverse Method.***

The coefficients of the 3 unknown variables were sorted into 3 by 3 cells to form a 3 by 3 matrix (cells D4 to F6). The RHS of the equation was then placed into 3 by 1 matrix cells (G4 to G6).

Cells E8 to E10 were highlighted to input the required formula into the

Q1

Ready

In SCILAB, the problem was solved as shown below

```
// Specie Flow Rates (m3/s)
Q_ab = 40;
Q_ac = 80;
Q_ba = 60;
Q_bc = 20;
Q_c_out = 150;
m_c_in = 195;
m_a_in = 1320;

// From the mass continuity system of equations
A = [120 -60 0;40 -80 0;80 20 -150];
B = [1320;0;-195];

// Values of xA, xB, xC
x = inv(A)*B

disp(x);
```

SECTION 2

Part A

The operation of a Resistance Temperature Detector (RTD) is based on the fact that the electrical resistance of the detector material changes with temperature. For Nickel, which is sometimes used in such detectors, the resistance, RT, at temperature T(°C) as a function of temperature is given by:

$$R_T = R_0 (1 + AT + BT^2 + CT^4 + DT^6)$$

Where R0 is the resistance of the detector at 0°C and A = 5.485×10^{-3}, B = 6.65×10^{-6}, C = 2.805×10^{-11}, D = -2×10^{-17} are constants. Consider a detector with R_0=100Ω and determine the temperature when its resistance is 300Ω.

Part B

The Van Der Waals equation gives a relationship between the pressure P (in atm), Volume (in L), and Temperature (in K) of a real gas:

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

Where n is the number of moles, R = 0.08026 L atm / mole K is the gas constant, and a (L^2 atm/ $mole^2$ and b (in L/mole) are material constants.

Consider 1.5 moles of Nitrogen (a = $1.39 L^2 atm/mole^2$, b= 0.03913 L/mole) at 25 ^oC stored in a pressure vessel. Determine the volume of a vessel if the pressure is 13.5 atm.

Repeat the determination of the volume for 15%, 30%, 45%, 60%, 75%, 90%, and 105% above the base pressure. Tabulate the pressure against the volume and plot. Solve this problem using

- The excel spreadsheet
- SCILAB.

In both cases, use any in-built functions to solve the problem.

2.1 Manual Calculation

Part A

Given, $R_T = R_0 (1 + AT + BT^2 + CT^4 + DT^6)$

where,

A = 5.485×10^{-3}

B = 6.65×10^{-6}

C = 2.805×10^{-11}

D = -2×10^{-17}

R_0 = resistance of the detector at 0^0C = 100Ω

Determine T when R_T = 300Ω.

2.1.1 Newton-Raphson Method

Firstly, we equate the given expression to 0.

$$f(T) = R_0(1 + AT + BT^2 + CT^4 + DT^6) - R_T = 0$$

The Newton-Raphson Equation is of the form: $T_{n+1} = T_n - \frac{f(T)}{f'(T)}$

Substituting R_T and R_0 into the equation,

$$f(T) = (AT + BT^2 + CT^4 + DT^6 - 2)$$

$$f'(T) = A + 2BT + 4CT^3 + 6DT^5$$

Substituting A to D,

$$f(T) = -2 + (5.485 \times 10^{-3})T + (6.65 \times 10^{-6})T^2 + (2.805 \times 10^{-11})T^4 - (2 \times 10^{-17})T^6$$

$$f'(T) = (5.485 \times 10^{-3}) + (13.3 \times 10^{-6})T + (11.22 \times 10^{-11})T^3 - (12 \times 10^{-17})T^5$$

Using the Newton-Raphson Equation and a calculator,

Guessing T_0 = 500 ^{o}C,

T_1 = 328.396921 ^{o}C

T_2 = 267.0884727 ºC

T_3 = 260.2914374 ºC

T_4 = 260.2186911 ºC

T_5 = 260.2186829 ºC

T_6 = 260.2186829 ºC = 260.22 ºC.

The value of T converged after 6 iterations.

2.1.2 Interval Bisection Method

Using the same equation and parameters as above,

$$f(T) = R_0(1 + AT + BT^2 + CT^4 + DT^6) - R_T = 0$$

Taking initial guesses a = 250 ºC and b = 290 ºC, we shall obtain c = a+b /2, f(a), f(b) and f(c) and iterate until a and b converge.

The following table was obtained from the calculations made:

a	**b**	**c = a+b/2**	**f(a)**	**f(b)**	**f(c)**
250	290	270	-0.1084	0.3364	0.1071
250	270	260	-0.1084	0.1071	-0.0024
260	270	265	-0.0024	0.1071	0.0519
260	265	262.5	-0.0024	0.0519	0.0247
260	262.5	261.25	-0.0024	0.0247	0.0111
260	261.25	260.625	-0.0024	0.0111	0.0044
260	260.625	260.3125	-0.0024	0.0044	0.0010
260	260.3125	260.15625	-0.0024	0.0010	-0.0007
260.15625	260.3125	260.234375	-0.0007	0.0010	0.0002
260.15625	260.234375	260.1953125	-0.0007	0.0002	-0.0003
260.1953125	260.234375	260.21484375	-0.0003	0.0002	-0.00004
260.21484375	260.234375	260.224609375	-0.00004	0.0002	0.00006
260.21484375	260.224609375	260.2197265625	-0.00004	0.00006	0.00001

From the table we can see that the value of our root lies between **260.2148 ºC** and **260.2246ºC**, which certifies the solution gotten from the Newton-Raphson Method.

2.2 Software Calculation

Part A

In Excel

NEWTON-RAPSHON METHOD

Given,
f(T) = AT + BT^2 + CT^4 + DT^6 - 2
f'(T) = A + 2BT + 4CT^3 + 6DT^5

Newton-Raphson Equation:

$$T_{n+1} = T_n - \frac{f(T)}{f'(T)}$$

Parameters	
A	0.005485
B	0.00000665
C	2.805E-11
D	-2E-17

Iteration	***Tn***	***f(T)***	***f'(T)***	***Tn+1***
1	500.0000	3.8456	0.0224	328.3969
2	328.3969	0.8196	0.0134	267.0885
3	267.0885	0.0748	0.0110	260.2914
4	260.2914	0.0008	0.0108	260.2187
5	260.2187	0.0000	0.0108	260.2187
6	260.2187	0.0000	0.0108	260.2187

Description of Process

- The polynomial function and its first derivative were first expressed in cells B3 and B4 respectively to enable easy typing of the formula. The expression for the Newton-Raphson equation is also depicted in the text box.
- Parameters A to D were evaluated in cells F3 to F6.
- In the C column, underneath Tn, an initial guess of 500°C was made and f(T) was calculated using the formula *"=(F4*C12) + (F5*(C12^2)) + (F6*(C12^4)) + (F7*(C12^6)) - 2"*.

- f'(T) was also evaluated using *"=F4 + (2*F5*C12) + (4*F6*(C12^3)) + (6*F7*(C12^5))"*.
- The Newton-Raphson equation was then evaluated in column F (Tn+1) using the formula *"=C12 - (D12/E12)"*.
- Finally, further iterations were made by putting the computed values of Tn+1 into Tn and resolving until the value of Tn converged at 260.2187^{o}C.

In SCILAB, the problem was solved as shown below

```
clc;

disp ("Newton Raphson Method")

// Given,
a = 5.485*10^-3;
b = 6.65*10^-6;
c = 2.805*10^-11;
d = -2*10^-17;

function y=f(T)
    y = a*T + b*T^2 + c*T^4 + d*T^6 - 2;
endfunction

function z=g(T)
    z = a + 2*b*T + 4*c*T^3 + 6*d*T^5;
endfunction

T0 = 0
for i=1:8
    T = T0 - f(T0)/g(T0)
    T0 = T
    disp ([i,T])
end

disp ("The root of the equation after 8 iterations is", T)
```

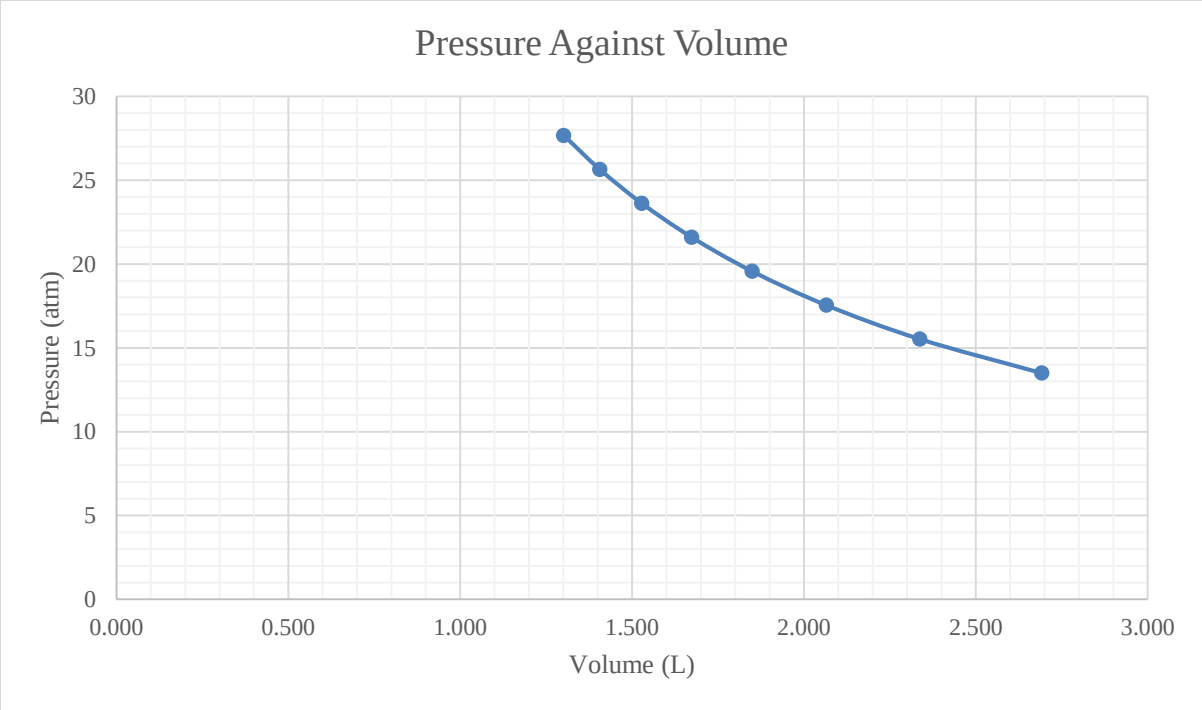
Part B

Description of Process

On Excel, the cubic equation was solved using ***Goal Seek***.

- The Van der Waals equation was rewritten into a cubic equation and equated to 0.

- The parameters for the equation were then typed into their individual cells C8 to C12 and the values for pressure were evaluated in cells A15 to A22.
- The values of P were evaluated in cells B15 to B22 for 15, 30, 45, 60, 75, 90, and 105% above the base pressure (13.5atm). e.g = 0.15*B15.
- Under "**Equation**", the formulas for the cubic equation were typed in for the various values of **P** and adjusted variable **V**.
- Goal Seek was selected from ***Data>Data Tools>What-If Analysis***.
- One after the other, cells C15 to C22 were *set to value* "**0**" *by changing cells* D15 to D22.
- The answers for Volume were then computed and displayed in the spreadsheet.
- An x-y scatter chart was inserted into the sheet and the graph of pressure against volume was plotted. The plot (shown below) showed proof of Boyle's Law where increase in pressure led to decrease in the volume of gas {$P = k/V$}.

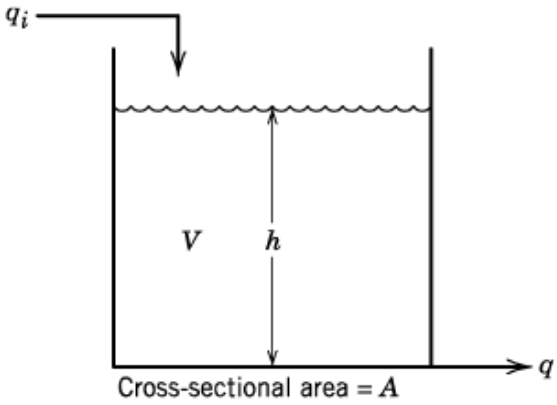


In SCILAB, the problem was solved as shown below

```
disp('Question 2B, Group 4')

//Variables given:
P = 13.5; //atm
R = 0.08206; //(L atm)/(mole K)
n = 1.5; //mole
a = 1.39; //L^2.atm/ mole^2
b = 0.03913; //L/mole
T = 298; //K == 25oC
V0 = n*R*T/P;

//The function.
function [y]=f(V)
    y = P*(V^3) - (P*n*b + n*R*T)*(V^2) + ((n^2)*a)*V -((n^3)*b*a);
endfunction

disp('For base pressure Pb = 13.5atm, the value of V is:')
P = 13.5;
Vb = fsolve(V0,f)
disp(Vb)

disp('For 15% above base pressure, P15 = 15.525atm, the value of V is:')
P = 15.525;
V15 = fsolve(V0,f)
disp(V15)

disp('For 30% above base pressure, P30 = 17.55atm, the value of V is:')
P = 17.55;
V30 = fsolve(V0,f)
disp(V30)

disp('For 45% above base pressure, P45 = 19.575atm, the value of V is:')
P = 19.575;
V45 = fsolve(V0,f)
disp(V45)

disp('For 60% above base pressure, P60 = 21.6atm, the value of V is:')
P = 21.6;
V60 = fsolve(V0,f)
disp(V60)

disp('For 75% above base pressure, P75 = 23.625atm, the value of V is:')
P = 23.625;
V75 = fsolve(V0,f)
disp(V75)

disp('For 90% above base pressure, P90 = 25.65atm, the value of V is:')
P = 25.65;
V90 = fsolve(V0,f)
disp(V90)

disp('For 105% above base pressure, P105 = 27.675atm, the value of V is:')
P = 27.675;
V105 = fsolve(V0,f)
disp(V105)
```

SECTION 3

Consider the schematic of the liquid tank system shown below where the inlet and outlet flows are volumetric quantities.



(a) Derive a model of the system relating the volumetric inlet flow to the liquid level in the tank.

(b) What type of model do you have?

(c) If the cross-sectional area of the tank is 100cm2 and the initial steady inflow to the tank is 10cm3 /s, find the initial steady-state liquid level in the tank.

(d) If the operator of the tank suddenly changed the inlet flow from the initial steady state by 50% and maintained it at the new value, use linear analysis method to determine a closed-form expression for the dynamic response of the liquid level to the operator command. 4

(e) For the same input as in (d), use the explicit Euler's method to determine dynamic response of the system using the following step lengths: 0.01, 0.05, 0.1, 0.5, 1, 10 using SCILAB. Plot the system response for the different values of step lengths.

(f) For the same input as in (d), use the Runge-Kutta 4th -order method to determine dynamic response of the system using the following step lengths: 0.01, 0.05, 0.1, 0.5, 1, 10 using SCILAB. Plot the system response for the different values of step lengths.

(g) Comment/discuss the results obtained in (e) and (f).

3.1 Manual Calculation

Assumptions

- The liquid density is constant
- The tank is cylindrical in shape with
- There is no generation nor consumption in the system

Let

$$A = Cross\ sectional\ area$$

$$V = Volume\ of\ Liquid\ in\ the\ tank$$

$$h = Liquid\ Level$$

$$q_i = Inlet\ Volumetric\ Flowrate$$

$$q = Outlet\ Volumetric\ Flowrate$$

Taking material balance around the tank,

$$Accumulation\ of\ Mass = Outlet\ mass\ flowrate - Outlet\ mass\ flowrate \quad \text{-------3.1}$$

$$Mass\ flow\ rate = Volumetric\ flowrate \times density$$

$$Initial\ molar\ flow\ rate = q_i \times \rho$$

$$Outlet\ molar\ flow\ rate = q \times \rho$$

$$Accumulation = -\frac{dM}{dt}$$

Equation 3.1 becomes

$$\frac{dM}{dt} = \rho q_i - \rho q \qquad \text{----3.2}$$

$$But\ the\ Mass\ of\ liquid\ in\ the\ tank\ =\ liquid\ Density \times Volume\ of\ the\ liquid\ in\ the\ tank$$

$$M = \rho V$$

$$\frac{dM}{dt} = \frac{d(\rho V)}{dt} = \rho q_i - \rho q$$

$$\frac{dM}{dt} = \frac{d(\rho V)}{dt} = \rho q_i - \rho q$$

Since the liquid density is assumed to be constant, $V = Ah$ where h is the liquid level

$$\frac{\rho\, d(V)}{dt} = \rho q_i - \rho q$$

$$\frac{\rho\, d(Ah)}{dt} = \rho q_i - \rho q$$

$$\frac{d(Ah)}{dt} = q_i - q$$

$$\frac{A\, d(h)}{dt} = q_i - q \qquad \text{----3.3}$$

For a liquid storage system, we have that $q = C_v\sqrt{h}$

$$\frac{A\, d(h)}{dt} = q_i - C_v\sqrt{h} \qquad \text{----3.4.}$$

b) The type of model generated in the equation 3.4 above is a lumped parameter model with an equation in the first order.

c) For a cross-sectional area of 100cm^2, and initial steady inflow to the tank is 10cm^3/s. The initial steady state liquid level in the tank.

$$A = 100cm^2, q_i = \frac{10cm^3}{s}, h = ?$$

From equation 3.3 derived above,

$$\frac{A\, d(h)}{dt} = q_i - C_v\sqrt{h}$$

At steady state, accumulation = $\frac{A\, d(h)}{dt} = 0$

$$0 = q_i - C_v\sqrt{h}$$

$$q_i = C_v\sqrt{h}$$

$$\sqrt{h} = \frac{q_i}{C_v}$$

This gives

$$h = \left(\frac{q_i}{C_v}\right)^2$$

Where C_v is the valve constant and it varies or flow coefficient and it varies for different equipment.

However, since the equipment wasn't specified, **Cv value is assumed to be 2** (A value well in the range of most equipment).

$$h = \left(\frac{10}{2}\right)^2$$

$$\boldsymbol{h = 25\ cm}$$

d) Assuming the change is a 50% increase, $q_{i\ new} = 150\%\ of\ q_i$

$$q_{i\ new} = 150\%\ (10) = 15\,\frac{cm^3}{s}$$

From equation 3.3

$$\frac{A\ d(h)}{dt} = q_i - C_v\sqrt{h}$$

The general form of a non-linear, one variable ODE is

$$\frac{dh}{dt} = f(h)$$

The linear analysis method can be used to solve. This gives

$$\frac{100\ d(h)}{dt} = 15 - 2\sqrt{h}$$

$$f(h) = \frac{d(h)}{dt} = 0.15 - 0.02\sqrt{h} \quad \text{----3.5}$$

At steady state, $h = \bar{h} = 25$

$$\frac{d(h)}{dt} = 0.15 - 0.02\sqrt{25}$$

$$\frac{d(h)}{dt} = f(\bar{h}) = 0.05$$

The equation $f(h) = \frac{d(h)}{dt} = 0.15 - 0.02\sqrt{h}$ can be linearised by Taylor series expansion.

The differential of equation 3.5 is given below:

$$f^{1}(h) = f(\bar{h}) + \left.\frac{df}{dh}\right|_{\bar{h}} (h - \bar{h})$$

$$\frac{dh}{dt} = 0.05 + (h - \bar{h}) \times \left[-0.02 \times \frac{1}{2\sqrt{h}}\right]$$

$$\frac{dh}{dt} = 0.05 + (h - 25) \times \left[-0.02 \times \frac{1}{2\sqrt{25}}\right]$$

$$\frac{dh}{dt} = 0.05 + (h - 25) \times [-0.002]$$

$$\frac{dh}{dt} = 0.05 - 0.002h + 0.05$$

$$\frac{dh}{dt} = -0.002h + 0.1$$

We may use Laplace transforms to solve the ODE

$$L\left[\frac{dh}{dt}\right] = L[-0.002h] + L[0.1]$$

$$s\,L(h) - L(0) = -0.002L(h) + \frac{0.1}{s}$$

$$L(0) = initial\ steady\ state\ liquid\ level = 25\ cm$$

$$s\,L(h) - 25 = -0.002L(h) + \frac{0.1}{s}$$

By collecting like terms

$$s\,L(h) + 0.002L(h) = \frac{0.1}{s} + 25$$

$$L(h)[s + 0.002] = \frac{0.1}{s} + 25$$

$$L(h) = \frac{\left(\frac{0.1}{s} + 25\right)}{[s + 0.002]}$$

$$L(h) = \frac{(0.1)}{s(s+0.002)} + \frac{25}{(s+0.002)}$$

Solving $\frac{(0.1)}{s(s+0.002)}$ by partial fractions gives $\frac{50}{s} - \frac{50}{(s+0.002)}$

$$\therefore L(h) = \frac{50}{s} - \frac{50}{(s+0.002)} + \frac{25}{(s+0.002)} = \frac{50}{s} - \frac{25}{(s+0.002)}$$

Then to find *h(t)*, the Laplace inverse of each term is taken.

$$L^{-1}(h) = h(t) = L^{-1}\left(\frac{50}{s}\right) - L^{-1}\left[\frac{25}{(s+0.002)}\right]$$

$$\boldsymbol{h(t) = 50 - 25e^{-0.002t}} \qquad \textbf{----3.5}$$

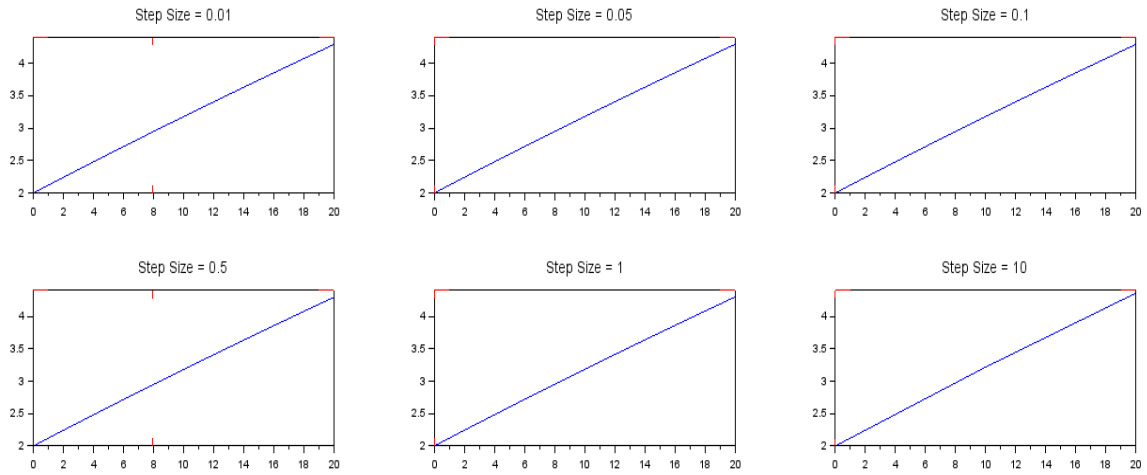
Equation 3.4 above presents a closed-form expression for the dynamic response of the liquid level to the operator command.

3.2 Software Calculation

e) Using the explicit Euler method on SCILAB to determine the dynamic response of the system to step input change in length gave a solution as provided below:

```
// Explicit Euler's Method

function h=Euler(f, step_size, t, y0)
    num_of_points = length(t);
    h = zeros(1, num_of_points);
    h(1) = y0;
    for i=1:num_of_points-1
        h(i+1) = h(i) + f(t(i), h(i))*step_size;
    end
endfunction

function dh_dt=f(t, h)
    Qi = 10;
    A = 100;
    Cv = 2;
    dh_dt = ((0.5*Qi)+Qi)/A - Cv/A*sqrt(h);
endfunction

step_sizes = [0.01, 0.05, 0.1, 0.5,1, 10]
i = 0
    clf
for step_size=step_sizes
    i = i+1
    t = 0:step_size:20
    h = Euler(f, step_size, t, 2)
    subplot(3, 3, i)
    plot(t, h)
    title("Step Size = " + string(step_size))
end
```

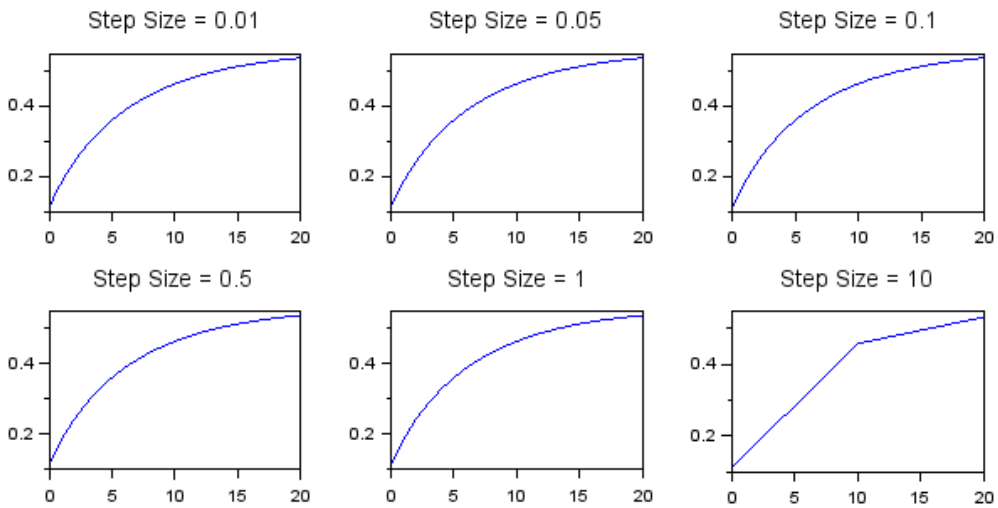
The plot of the system response is presented below:



f) Using the same input as in question d, we solve using Runge-Kutta 4th order method to determine the response of the system. The solution is as shown below:

```
// Runge-Kutta Method

function h=Runge_Kutta(f, step, h0, t)
    N = length(t)
    h = ones(1, N)
    h(1) = h0

    for i=1:N-1
        k1 = f(t(i), h(i))
        k2 = f(t(i) + step/2, h(i) + step/2*k1)
        k3 = f(t(i) + step/2, h(i) + step/2*k2)
        k4 = f(t(i)+step, h(i)+step*k3)

        h(i+1) = h(i) + step/6 * (k1 + 2*k2 + 2*k3 + k4)
    end
endfunction

Qi = 10;
A = 100;
Cv = 20;
h0 = (10/k)^2;

function dh_dt=f(t, h)
    dh_dt = ((0.5*Qi)+Qi)/A - Cv/A*sqrt(h)
endfunction

step_sizes = [0.01, 0.05, 0.1, 0.5,1, 10]
i = 0
clf

for step_size=step_sizes
    i = i+1
    t = 0:step_size:20
    h = Runge_Kutta(f, step_size, h0, t)
    subplot(3, 3, i)
    plot(t, h)
    title("Step Size = " + string(step_size))
end
```

The system response at the step lengths of 0.01, 0.05, 0.1, 0.5, 1, and 10 were plotted on SCILAB and the results are presented below.



g) The plots from the step lengths in the e parts of the question gave straight lines graphs. From the graphs, the value with the smallest step size was expected to get the most accurate value and from the graphs in e above, the most accurate is the step length of 0.01.

in the f part, unlike in the previous linear plot of k obtained from Euler's equation, this is as a result of truncating to the order of four which takes account of the variations in space and other parameters.

SECTION 4

The following are measurements of the rate coefficient, k, for the reaction below at different temperatures, T.

$$CH_4 + O \rightarrow CH_3 + OH$$

T(K)	595	623	761	849	989	1076	1146	1202	1382	1445	1562
$Kx10^{20}$	2.12	3.12	14.4	30.6	80.3	131	186	240	489	604	868

(a) Use the method of least-squares to best fit to the data a function of the form

$$ln(k) = C + b\ ln(T) - \frac{D}{T}$$

(b) Usually the rate coefficient is expressed in the form of an Arrhenius equation $k = AT^b e^{\frac{-Ea}{RT}}$ where A and b are constants, R= 8.134 J/moleK is the universal gas constant and Ea is the activation energy for the reaction. Having determined the constants C,b and D in part a, deduce the values of A (m3/s) and Ea (J/mole) in the Arhenius equation.

(c) Formulate the appropriate least squares problem and solve the problem using

(i) A calculator

(ii) Spreadsheet

(iii) SCILAB

4.1 Manual Calculation

In(T)	1/T	In(K)	$(InT)^2$	$(\frac{1}{T})^2*10^{-6}$	(I/T)ln(K)	In(T)In(K)	In(T)1/T
6.3386	0.00168	-45.3003	40.8142	2.8224	-0.0761	-289.406	0.01073
6.4345	0.00161	-44.9139	41.4028	2.5921	-0.0723	-288.999	0.01036
6.6346	0.00131	-43.3845	44.0179	1.7161	-0.0568	-287.839	0.00869
6.7441	0.00178	-42.6307	45.4829	3.1684	-0.0759	-287.506	0.01200
6.8967	0.00101	-41.6659	47.5645	1.0201	-0.0421	-287.357	0.00697
6.9810	0.00093	-41.1765	48.7344	0.8649	-0.0383	-287.453	0.00649
7.0440	0.00087	-40.8260	49.6179	0.7569	-0.0355	-287.578	0.00613
7.0917	0.00083	-40.5710	50.2922	0.6889	-0.0337	-287.717	0.00589
7.2313	0.00072	-39.8593	52.2917	0.5184	-0.0287	-288.235	0.00521
7.2759	0.00069	-39.6481	52.9387	0.4761	-0.0274	-288.476	0.00502
7.3537	0.00064	-39.2855	54.0769	0.4096	-0.0251	-288.894	0.00471
∑In(T)=	∑1/T=	∑In(K)=	$\sum(InT)^2$ =	$\sum(1/T)^2$ =	∑1/Tin(K) =	∑InT.InK =	∑InT(1/T) =
76.0761	0.01207	-459.262	527.2341	$15.0339*10^{-6}$	-0.5119	-3169.459	0.0822

From

$$\ln K = c + b \ln T - \frac{D}{T} \qquad \text{-----4.0}$$

It can be deduced that for multiple iterations to obtain values of k at corresponding temperatures,

$$\sum \ln K = nc + b \sum \ln T + D_1 \sum \frac{1}{T} \qquad \text{-----4.1}$$

For simplification, -D is replaced with the term $D_1 = -D$ used here

To evaluate using the least squares method, there is a need to develop multiple equations to solve simultaneously. By multiplying the equation 4.1 by various functions of T, relationships combining the three variables may be created and analysed.

Multiplying through by ln T gives

$$\sum \ln K \ln T = c \sum \ln T + b \sum (\ln T)^2 + D_1 \sum \frac{1}{T} \ln T \qquad \text{-----4.2}$$

Multiplying through by 1/T gives

$$\sum \ln K \frac{1}{T} = c \sum \frac{1}{T} + b \sum \ln T \frac{1}{T} + D_1 \sum \left(\frac{1}{T}\right)^2 \qquad \text{-----4.3}$$

Three equations that may be solved simultaneously have been generated.

$$\sum \ln K = nc + b \sum \ln T + D_1 \sum \frac{1}{T} \qquad \text{-----4.1}$$

$$\sum \ln K \ln T = c \sum \ln T + b \sum (\ln T)^2 + D_1 \sum \frac{1}{T} \ln T \qquad \text{-----4.2}$$

$$\sum \ln K \frac{1}{T} = c \sum \frac{1}{T} + b \sum \ln T \frac{1}{T} + D_1 \sum \left(\frac{1}{T}\right)^2 \qquad \text{-----4.3}$$

From the table above,

$\sum \ln K = -459.262$, $\sum \ln T = 76.0761$, $\sum \frac{1}{T} = 0.01207$, $\sum \ln K \ln T = -3169.459$,

$\sum (\ln T)^2 = 527.2341$, $\sum \ln K \frac{1}{T} = -0.5119$, $\sum \frac{1}{T} = 0.01207$, $\sum \ln T \ \frac{1}{T} = 0.0822$, $\sum \left(\frac{1}{T}\right)^2 =$

15.0339×10^{-6}, n = number of iterations = 11

$$-459.262 = 11C + 76.0761b + 15.0339D_1 \qquad \text{-----4.1}$$

$$-3169.459 = 76.0761C + 527.2341b + 0.0822D_1 \qquad \text{-----4.2}$$

$$-0.5119 = 0.01207C + 0.0822b + (15.0339 \times 10^{-6})D_1 \qquad \text{-----4.3}$$

In Matrix form equations 1-3 can be presented as:

$$\begin{bmatrix} 11 & 76.0761 & 15.0339 \\ 76.0761 & 527.2341 & 0.0822 \\ 0.01207 & 0.0822 & (1.50339 \times 10^{-6}) \end{bmatrix} \begin{bmatrix} c \\ b \\ D_1 \end{bmatrix} = \begin{bmatrix} -459.262 \\ -3169.459 \\ -0.5119 \end{bmatrix}$$

$$Ax = B$$

$$A = \begin{bmatrix} 11 & 76.0761 & 15.0339 \\ 76.0761 & 527.2341 & 0.0822 \\ 0.01207 & 0.0822 & (1.50339 \times 10^{-6}) \end{bmatrix}, x = \begin{bmatrix} c \\ b \\ D_1 \end{bmatrix}, B = \begin{bmatrix} -459.262 \\ -3169.459 \\ -0.5119 \end{bmatrix}$$

Solving using crammer's rule we have,

$$c = \frac{Dc}{DetA} \qquad \text{-------4.4a}$$

$$b = \frac{Db}{DetA} \qquad \text{-------4.4b}$$

$$D_1 = \frac{DD}{DetA} \qquad \text{-------4.4c}$$

Where DetA is the determinant of the coefficient matrix A and

Dc, Db, DD are the determinants of the matrices of $c, b\ and\ D$ as defined below.

$$Dc = \begin{vmatrix} -459.262 & 76.0761 & 15.0339 \\ -3169.459 & 527.2341 & 0.0822 \\ -0.5119 & 0.0822 & (1.50339 \times 10^{-6}) \end{vmatrix} = -3.0019 \times 10^{-4}$$

$$b = \begin{vmatrix} 11 & -459.262 & 15.03390 \\ 76.0761 & -3169.459 & 00.0822 \\ 0.01207 & -0.5119 & (1.50339 \times 10^{-6}) \end{vmatrix} = -0.13919$$

$$D_1 = \begin{vmatrix} 11 & 76.0761 & -459.262 \\ 76.0761 & 527.2341 & -3169.459 \\ 0.01207 & 0.0822 & -0.5119 \end{vmatrix} = 0.01566$$

$$Det\ A = \begin{vmatrix} 11 & 76.0761 & 15.0339 \\ 76.0761 & 527.2341 & 0.0822 \\ 0.01207 & 0.0822 & (1.50339 \times 10^{-6}) \end{vmatrix} = 9.886 \times 10^{-6}$$

$$c = \frac{-3.0019 \times 10^{-4}}{9.886 \times 10^{-6}} = \mathbf{-30.365}$$

$$b = \frac{-0.13919}{9.886 \times 10^{-6}} = \mathbf{-1.408 \times 10^{-8}}$$

$$D_1 = \frac{0.01566}{9.886 \times 10^{-6}} = \mathbf{1584.20}$$

$$\therefore \ln K = \mathbf{-30.365 - 1.408 \times 10^{-8}} \ln T - \frac{\mathbf{1584.20}}{T}$$

b) The Arrhenius equation is given by $\boldsymbol{k = AT^b e^{\frac{-Ea}{RT}}}$ and R = 8.134 J/mol.K

Taking ln to both sides,

$$\ln K = \ln A + b \ln T - \frac{Ea}{RT}$$

This is similar to equation 4.1 $\ln K = c + b \ln T - \frac{D}{T}$ and since $D_1 = -D$, the equation becomes,

$$\ln K = \mathbf{-30.365 - 1.408 \times 10^{-8}} \ln T - \frac{\mathbf{1584.20}}{T}$$

On comparing,

$$ln\ A = -30.365$$

$$A = e^{-30.365}$$

$$\boldsymbol{A = 6.496 \times 10^{-14}}$$

Also,

$$\frac{Ea}{RT} = \frac{1584.20}{T}$$

$$\frac{Ea}{R} = 1584.20$$

$$Ea = \mathbf{1584.20}\boldsymbol{R} = \mathbf{1584.20(8.314) = 13171.04}\ \boldsymbol{J/mol}$$

4.2 Software Calculation

On Excel, Solver was used to evaluate the value of the constants c, b and D.

L	M	N	O	P	Q
	Variable Coefficients from the Least Square equations				
		C	b	D	
	EQUATION 4.1	11	76.07615	0.011479	
	EQUATION 4.2	76.07615	527.2348	0.078211	
	EQUATION 4.3	0.011479	0.078211	1.3267E-05	
	Using Solver to get the left hand sides equal to the right hand sides of of equations 4.1-4.3 and to get the values of c, b and D				
		R.H.S		L.H.S	
	EQUATION 4.1	-459.2618	=	-459.26176	
	EQUATION 4.2	-3169.46	=	-3169.46	
	EQUATION 4.3	-0.48666	=	-0.48666	
	VARIABLE	c	b	D	
	VALUE	-55.41282	2.4974546	-3460.0079	
		A	8.6E-25		
		Ea	28766.506		

On SCILAB, the least squares method was used in matrix form. **B** was used to create an 11x3 matrix of ones. X1 and X2 were used to replace the second and third columns respectively. Then the constant c, b and D were represented by x and the disp command was used.

```
//The temperature and rate constant is given in the matrix form below
T=[595;623;761;849;989;1076;1146;1202;1382;1445;1562];
k=[2.12;3.12;14.4;30.6;80.3;131;186;240;489;604;868];
K=k*10**-20;
N=length(T);
//The natural log of the temperature and rate constant are given below
X1=log10(T)/log10(2.718281828);
Y=log10(K)/log10(2.718281828);
X2=1./T;
B=ones(N,3);
B(:,2:3)=[X1,X2];
x=inv(B'*B)*B'*Y;
disp("The values of the constants c, b, and D are gotten as", x)
disp ("respectively.")
c=x(1);
b=x(2);
D=x(3);
//comparing with the arrhenius equation
A=exp(c);
//GAS CONSTANT R=8.314J/MOLK
R=8.314;
Ea=R.*D;
disp('the value of A is '+string(A))
disp('the activation energy is '+string(Ea))
```

DISCUSSION

In this section, the results obtained from the 4 sections are discussed one after the other. It contains an overview of the results achieved, validation of said results, the challenges encountered in solving the problems, how they were overcome, and the lessons learnt.

Section 1

This problem saw a chemical engineering process application which involved three chemical reactors A, B, C, operating in steady state. The concentrations of the specie of interest in the different reactors (x_A, x_B, and x_C) were to be determined. This was achieved by carrying out material balance on each reactor to get the continuity equations for each reactor and solve for the unknown concentration variables simultaneously.

By writing the equations in the matrix form, the solution was computed manually using Crammer's Rule, the Matrix Inverse method, and the Gaussian Elimination method. The same problem was solved using computer software. The Spreadsheet application, Excel and programming app, SCILAB were used to achieve the results. The Matrix Inverse was the method used in obtaining the solution in both applications.

Validation of Results

All three manual methods achieved the same values of concentration (x_A, = 14.67, x_B = 7.33, and $x_C = 10.1$. The legitimacy of the results was corroborated by the software used. Both Excel and Scilab gave the same values for concentration in each reactor.

Challenges Encountered

No major challenge was encountered in the solution of this problem. However, at the inception of finding the solution manually, a random error in the continuity equation gave different values of xC. This error was discovered after a careful look through the entire solution and corrected.

Section 2

This question was in two parts.

Part A was based on the operation of a Resistance Temperature Detector (RTD) in which its electrical resistance changes with temperature. The resistance for Nickel was given as a function of temperature. The aim was to determine the temperature of the detector when its resistance was 300Ω. The function given was non-linear and had to be solved using the Newton-Raphson and Interval Bisection methods. The solution was obtained both manually and with computer software (Scilab and Excel).

Part B showed the Van der Waals relationship between the pressure, volume, and temperature for a real gas. The equation was rearranged into a cubic equation with volume, V acting as the independent variable. The volume of the gas was to be determined at base pressure of 13.5atm and multiple step increases in pressure using Excel and Scilab. Both cases were solved using built-in functions in the applications. For Scilab, the "fsolve" function was used to solve the cubic equation. In Excel, "Goal Seek" was used to determine the volume at different pressures.

Validation of Results

Part A

After manually iterating the Newton-Raphson equation six times, and taking an initial guess of 500°C, the temperature required converged at 260.22°C. The same temperature was achieved

after the same iterations were made using both SCILAB and Excel. The result was further confirmed using the Interval Bisection method manually. With initial guesses of "a" and "b" as 250°C and 290°C, the temperature fell in the range of 260.2148°C to 260.2246°C after 13 iterations.

Part B

The volumes were first computed manually and confirmed using computer software. A plot of P vs V was made and it showed the proof of Boyle's Law which states that the volume of a given mass of gas is inversely proportional to its pressure.

Challenges Encountered

No major challenges were faced.

Section 3

A model for a liquid tank system, relating its inlet volumetric flow rate to the height of liquid in the tank was to be derived. After derivation, the type of model obtained was a non-linear, lumped parameter model of first order. The assumptions made were that the liquid was of constant density and the flow was in steady-state.

The initial steady-state liquid level in the tank was calculated as 25cm, based on a key assumption and using the equation derived. A closed-form expression for the dynamic response of the liquid level to a change in operation was obtained by first linearising the model equation using Taylor Series Expansion, and solving the resulting ordinary differential equation (ODE) using Laplace Transforms.

This response to step change was then shown in different plots for various step lengths in SCILAB. The methods used were Euler's Explicit method and the Runge-Kutta 4th Order method.

Validation of Results

The initial model obtained was a non-linear equation of first order. By linearizing the equation, the plots obtained for the different step inputs were linear for the Euler method; showing a directly proportional relationship between the step input, t and liquid level, h. Using the Runge-Kutta 4th order method, the relationship was not linear as the liquid level increased exponentially from step input changes. This graph is in reality more accurate than the Euler's method since it more accurately accounts for a the spatial and other differences in the system.

Challenges Encountered

There was difficulty in understanding the question regarding the change in response of the system and its interpretation into SCILAB code. Through group discussions, collaborations with other groups, studying the slides and textbooks, and YouTube tutorials, more understanding was gained in writing the code and plotting the graphs on SCILAB.

Section 4

Measurements for the rate coefficient, k of a reaction at different temperatures, T were tabulated in the question. By using the method of least-squares, we were able to best fit a function of the form: In(k) = C + b In(k) - D/T to the data given. This was achieved by manually developing a system of simultaneous equations and solving them using Crammer's Rule. The values of C, b, and D were obtained as -30.365, -1.408×10^-8, and -1584.2 respectively.

We were also tasked to determine the activation energy, Ea and Arrhenius constant, A from the Arrhenius equation, $k = AT^b . e^{-\frac{E_a}{RT}}$. Ea and A were obtained as 13171.04J/mol and 6.496×10^-14 respectively.

Validation of Results

The results obtained from the least-squares method were also achieved using SCILAB and Excel. However, there was variation in the results gotten manually from the ones gotten using software.

Challenges Encountered

The answer obtained from manual computation was different from those calculated using SCILAB and Excel. This was due to approximation and perhaps random human errors. Also, the many steps needed to solve the problem is a possible source of recurring mistakes and errors. However, after careful consideration with the rest of the group, the final solution was achieved.

What We Learnt

The problems in this assignment ranged from the modelling of chemical engineering processes to solving various problems using Numerical Methods. The solutions to the problems were also achieved using programming software (SCILAB) and computer spreadsheets (Microsoft Excel). The knowledge gained from this part of the assignment was highly significant, increased our literacy in digital computing and allowed the members of the team work together to achieve a goal.

A summary of the lessons learnt are listed below:

- The cooperation and coordination of tasks to achieve solutions to engineering problems.

- Solving simultaneous equations using different numerical and computing methods (Crammer's Rule, Gaussian Elimination, Matrix Inverse, "MMULT" and "MINVERSE" Excel functions, writing and ordering of matrices in SCILAB)
- Solutions of linear and non-linear algebraic equations of different orders using the Goal Seek function in Excel, writing of code and formulas for Newton-Raphson iterations in SCILAB and Excel.
- Using the Interval Bisection method and writing the code in SCILAB.
- Linearization of non-linear differential equations using Taylor Series Expansion.
- How to write code for step input changes and plotting of graphs in SCILAB.

CONCLUSION

This assignment was done as an assessment to test and establish the knowledge of Chemical Engineering Analysis (CHG 433). Four questions given in four sections were evaluated using various numerical methods and the computer application software SCILAB and Microsoft Excel. The solutions were first computed manually, by making assumptions and applying theories. Then the required formulas and lines of code were typed into the software mentioned above.

Problem 1 depicted a chemical engineering process application involving 3 chemical reactors operating at steady state. By making material balance around each reactor, a mass continuity system of three simultaneous equations was generated. The equations were used to evaluate the concentrations of the required specie in each reactor: x_A, x_B, and x_C by solving the using Crammer's Rule, Gaussian Elimination, and Matrix Inverse. The Matrix inverse method was used in solving the system of equations on SCILAB and Microsoft Excel and the answers obtained were all the same.

Question 2 was in two parts. Part A dealt with the determination of the temperature in a Resistance Temperature Detector (RTD) operation. Using the Newton-Raphson and Interval Bisection methods, the temperature of the detector when its resistance was 300Ω was calculated as 260.22 ^{o}C. The Excel and SCILAB solutions also employed the Newton-Raphson equation to calculate the required temperature. In part B, the volume of a real gas was determined at different pressures by using the Van der Waals equation. The problem was solved in SCILAB with the "fsolve" function, and in Excel using "Goal Seek". Pressure was then plotted against volume and plotted in the Excel sheet. The plot showed an inverse relationship between pressure and volume (Boyle's Law).

A model was derived for a liquid tank system with inlet and outlet volumetric flow rates in question 3. The model related the volumetric inlet flow to the level of liquid in the tank. The model derived was a lumped parameter model. An expression for the dynamic response of the liquid level to change in operation commands was also determined by linearising the non-linear model equation. This was done using Taylor Series Expansion, and the resulting ODE was evaluated with Laplace Transforms. The solution of the resulting equation was taken to SCILAB to show the response of the system to different step lengths by using the Euler Explicit function and Runge-Kutta 4th order methods. The R-K graph is in reality more accurate than the Euler's method since it more accurately accounts for a the spatial and other differences in the system.

In question 4, the method of least-squares was used to best fit a function of the form $In(k) = C + bIn(T) - \frac{D}{T}$ to the data given. This data showed the measurements of the rate coefficient, k for a reaction at different temperatures, T. By manually developing a system of three simultaneous equations using least squares and Crammer's rule, the values of C, b, and D were obtained. The values computed manually varied from the ones gotten in the software due to approximation and possible random errors. From comparing the function and constants gotten to the log of the Arrhenius Equation, $k = AT^b . e^{-\frac{E_a}{RT}}$, the values of A and E_a were deduced.

All the problems were solved, typed, and formatted properly. The necessary plots, codes snippets, and Excel spreadsheet images and formulas are also included in the report.